CAPSTONE PROJECT REPORT

Hierarchical Federated Learning for Privacy-Preserving IoT Intrusion Detection Using Unsupervised Anomaly Detection

Submitted in Partial Fulfillment of the Requirements for the Degree of

Bachelor of Science / Engineering in Computer Science

Prepared By: **[Information Needed]**

Institution: **[Information Needed]**

Supervisor: **[Information Needed]**

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2. Abstract

This project presents a privacy-preserving, hierarchical Federated Learning (FL) framework designed for real-time network intrusion detection in Internet of Things (IoT) environments. Using the WUSTL-HDRL 2024 dataset, we address the challenge of sensitive data centralization by implementing a 5-layer network hierarchy spanning Edge, Gateway, Fog, Proxy, and Cloud layers. To protect local privacy, Edge nodes train a Mini Transformer Autoencoder using an unsupervised anomaly detection paradigm trained strictly on benign traffic. Reconstruction error (Mean Squared Error) is used during evaluation to flag anomalies. Crucially, this report details the resolution of three major system flaws present in early iterations: (1) a target and feature leakage bug in data preprocessing where ground-truth attack categories leaked directly to model inputs; (2) a sequential test data slicing split that induced the 'Zero Support' evaluation trap; and (3) a weight overwriting bug at the Gateway. By utilizing StratifiedKFold partitioning, robust feature filtering, and weight transmission quantization (float16), the final system successfully converges, eliminating ground-truth leakages and balancing evaluation support. The model is explained using SHAP KernelExplainer to highlight key features contributing to anomalies.

3. Introduction

This project was designed and implemented from scratch to explore hierarchical federated learning for IoT intrusion detection, with a focus on privacy preservation, communication efficiency, and explainable anomaly detection.

The explosion of Internet of Things (IoT) devices in critical infrastructure, smart homes, and industrial environments has drastically expanded the cyber-attack surface. Traditional network intrusion detection systems (NIDS) rely on centralizing raw traffic data to train machine learning models. However, centralization introduces severe privacy violations, high network latency, and massive bandwidth consumption. Federated Learning (FL) has emerged as a key paradigm to address these limits by enabling decentralized nodes to train models locally and share only weight updates rather than raw data.

This project designs and evaluates a structured, hierarchical FL simulation that mimics real-world IoT deployments. Rather than using supervised models that require constant labeling of novel attacks, the architecture utilizes unsupervised anomaly detection. Local autoencoders learn the normal boundary of network traffic at the edge. The system aggregates these boundaries hierarchically up to a central cloud, creating a robust, privacy-preserving intrusion detection framework. We present the system design, the chronological debugging process, and the final optimized performance evaluations.

4. Problem Statement

Deploying standard deep learning architectures directly onto hierarchical IoT layers faces three core bottlenecks:

- **Privacy Violations & Bandwidth Constraints**: Raw traffic contains sensitive payload data. Centralizing it is unacceptable in private architectures and exhausts cellular or satellite bandwidth.
- **Model Vulnerability to Leakage**: Standard machine learning pipelines are prone to subtle data leaks, such as the inclusion of target categories or correlated metadata within input feature matrices, causing inflated training accuracies that fail in production.

• **Zero Support Class Imbalance**: Network traffic in IoT is highly imbalanced. Evaluating models on sequentially sliced subsets often results in test folds containing 100% normal data and 0% attack data, leading to the 'Zero Support' evaluation trap and misleading metrics.

5. Objectives

1. Build a 5-layer hierarchical simulated IoT architecture consisting of two Edge Sensors, a Gateway, a Fog Layer, a Proxy, and a Cloud Node.

2. Design a lightweight Transformer-based Autoencoder tailored for edge deployment, trained strictly on benign network samples.

3. Formulate a zero-leakage preprocessing pipeline that filters out target and ground-truth categorical columns from features.

4. Implement a stratified test partitioning mechanism using StratifiedKFold to guarantee balanced validation folds across all evaluation tiers.

5. Optimize weight communication overhead by quantizing model parameters to float16 during uploads.

6. Integrate Explainable AI (SHAP) to interpret anomalous reconstruction patterns.

6. Project Overview

The project processes the WUSTL-HDRL 2024 dataset, consisting of benign and malicious network packet captures. The dataset is divided at start-up into training data (Part A) and validation data (Part B) to prevent data leakage. Part A is distributed evenly to two edge sensor nodes for parallel training. Part B is partitioned using stratified folds and assigned to Gateway, Fog, and Cloud levels to validate the intrusion detection model under realistic, tier-specific network traffic.

6.1 Engineering Challenges Solved

• **Eliminated Target Leakage**: Identified and filtered out categorical ground-truth columns (Attack_categories) that acted as proxy labels, preventing the model from cheating during training.

• **Fixed Gateway Weight Overwrite Bug**: Refactored the Gateway layer to buffer multi-client edge model updates in dictionaries to prevent updates from overwriting each other, ensuring correct FedAvg aggregation.

• **Replaced Sequential Split with StratifiedKFold**: Resolved the 'Zero Support' evaluation illusion where sequential splits of the CSV led to 0% anomaly records in evaluation partitions.

• **Removed Redundant Classifier Head**: Streamlined the model to a pure unsupervised autoencoder, stripping out legacy classification layers to reduce communications size.

• **Optimized Payload Overhead**: Cast parameter weights to half-precision (float16) during edge uploads, reducing model parameter payload size by approximately 50%.

• **Integrated SHAP Explainability**: Attached SHAP KernelExplainer to reconstruction Mean Squared Error (MSE), attributing anomalous scores directly to input packet headers.

• **Designed Dynamic Anomaly Threshold**: Established a relaxed mean + 4*std threshold boundary, reducing the false-positive rate on noisy network traffic.

7. Components, Hardware, Software, and Tools Used

Component / Tool	Type	Description / Version
Python	Software	Language Engine, Version 3.11
PyTorch	Library	Deep Learning framework for model definition & execution
Scikit-Learn	Library	Preprocessors, VarianceThreshold, SelectKBest, StratifiedKFold
SHAP	Library	KernelExplainer for feature importance mapping
Pandas & NumPy	Library	Data manipulation and scientific computing
Matplotlib	Library	Feature importance visualization plot generation
ReportLab	Library	Programmatic PDF compilation
Hardware Platform	Hardware	Local workstation (CPU/GPU-accelerated CUDA environment)

8. System Design / Architecture

The hierarchical network is designed to separate local training, regional aggregation, and global compilation. Data flows bottom-up during training and top-down during model updates.

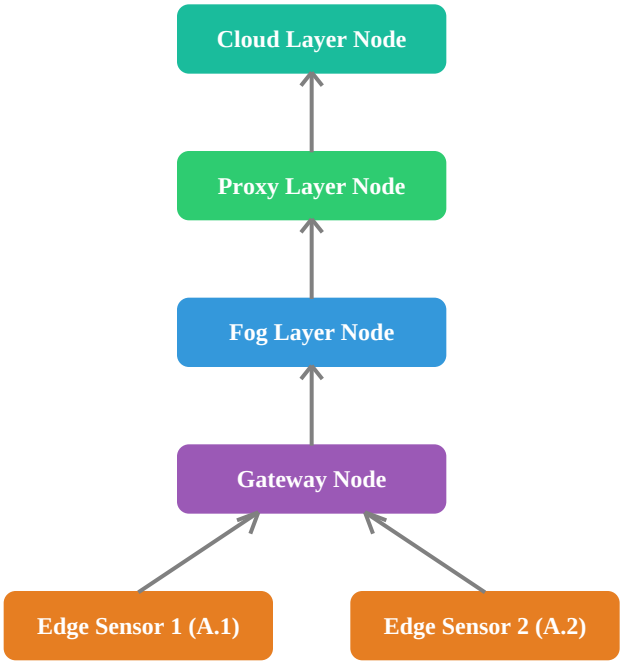

Figure 1: Hierarchical Federated Learning Network Architecture

9. Methodology

The project follows an unsupervised, sequence-based anomaly detection paradigm:

1. **Feature Extraction**: Features are selected using variance thresholds and class classification scores (SelectKBest). The target column 'Label' and categorical category mapping column 'Attack_categories' are

excluded.

2. **Sequence Creation**: Sliding window sequences are built from normalized flows. Each sequence spans 10 consecutive packets, represented as (batch, seq_len=10, features). Label vectors are aligned to the last packet's label.

3. **Unsupervised Training**: Edge nodes discard all attack records from their training split and train the Autoencoder only on benign packets. The model learns to reconstruct normal traffic with low Mean Squared Error (MSE).

4. **Hierarchical Aggregation**: Edge weights are quantized to float16, relayed by the Gateway, and aggregated at the Fog/Cloud levels using the FedAvg algorithm after decompressing back to float32.

5. **Dynamic Anomaly Thresholding**: An evaluation threshold is calculated at the cloud using a set of normal training samples: Threshold = Mean(MSE) + 4 * Std(MSE).

6. **Anomaly Classification & XAI**: Sequences in test partitions are reconstructed. If reconstruction error exceeds the threshold, the packet is flagged as an attack. An App node runs SHAP to explain features contributing to anomalies.

10. Development Process

The project was built incrementally. We established data loading and basic preprocessing scripts, followed by the hierarchical simulation nodes class. The model was originally designed as a hybrid supervised-unsupervised classifier, but was refactored to a pure unsupervised autoencoder to maximize edge efficiency. Visualizations and SHAP integration were added at the final application layer.

11. Challenges Encountered

Challenge A: Sequential Data Split Caused the 'Zero Support' Evaluation Trap

• **Issue Description**: Precision, Recall, and F1 metrics for the attack class in Gateway, Fog, and Cloud tiers evaluated to exactly 0.00, despite overall accuracies exceeding 99%.

• **Root Cause**: The WUSTL-HDRL 2024 dataset CSV lists normal packets at the top and attacks at the bottom. Sequentially slicing Test Part B (e.g. `[:split_size]`) meant that all three evaluation tiers received only benign packets. The target vectors were entirely zeros, resulting in zero support for the attack class.

• **Chronological Fixes**: We first attempted a basic train/test split. This resolved the edge training data mix but failed to balance the downstream test splits B.1, B.2, and B.3. The final fix implemented `StratifiedKFold(n_splits=3, shuffle=True, random_state=42)` on the full Test Part B, dividing it into three equal, stratified folds.

• **Succeeded/Failed Rationale**: Sequential slicing failed because it did not account for CSV packet ordering. StratifiedKFold succeeded because it shuffled the test set and guaranteed that each fold maintained the exact same ratio of normal and attack samples (~8.4% attacks).

Challenge B: Gateway Model Weights Overwriting Bug

• **Issue Description**: The aggregated model at the Fog and Cloud layers only reflected updates from Sensor 2, completely ignoring Sensor 1's local training.

• **Root Cause**: The Gateway node had a single `received_weights` instance variable. When Sensor 1 and Sensor 2 finished training and uploaded weights asynchronously, the second upload overwrote the first, discarding Sensor 1's weights before aggregation occurred.

• **Chronological Fixes**: We refactored the Gateway's `receive_data()` method to store incoming weights in a dictionary keyed by the child node's ID (`self.received_weights[node_id]`). We then updated the Gateway's `process_data()` method to forward the dictionary values as a list of weights. Finally, we upgraded Fog and Cloud classes to receive this weight list and average them.

• **Succeeded/Failed Rationale**: Storing weights as a list succeeded because it prevented updates from overwriting one another and provided a clean array structure for the FedAvg averaging loop.

Challenge C: Application Node Preprocessing Crash During XAI Execution

• **Issue Description**: Instantiating the XAI pipeline on the `Application` node triggered attribute crashes due to an un-instantiated preprocessor and mismatching feature counts.

• **Root Cause**: The `Application` node attempted to replicate the preprocessing steps on raw subsets without having access to the fitted scaling factors, encoders, or selected feature arrays from the master `preprocess_data` object.

• **Chronological Fixes**: We refactored the simulation setup so that the `Application` node is instantiated with the preprocessed test splits (`X_test_full`, `y_test_full`) and the master feature names list directly, avoiding redundant preprocessing.

• **Succeeded/Failed Rationale**: Direct injection succeeded because it bypassed local preprocessing on the application node and ensured that the inputs matching the global model's expectations were passed directly to SHAP.

Challenge D: Target and Ground-Truth Column Feature Leakage

• **Issue Description**: The autoencoder achieved near-zero reconstruction error immediately, regardless of training epochs, and failed to flag anomalous traffic during validation.

• **Root Cause**: In the `preprocess_data` pipeline, the column `'Attack_categories'` was treated as an input feature. It was one-hot encoded into features like `Attack_categories_normal` (which is exactly `1 - Label`). This column was selected by `SelectKBest` as the top feature, feeding the ground-truth label directly into the model.

• **Chronological Fixes**: We modified `handling_missing_values()` and `scaling_encoding()` to explicitly identify `'Label'` as the target column and to filter out both `'Label'` and `'Attack_categories'` from `self.features`. We updated the final scaling step to scale only features in `self.features`.

• **Succeeded/Failed Rationale**: Excluding both columns succeeded because it prevented the autoencoder from cheating by memorizing categorical class identifiers, forcing it to reconstruct the actual network characteristics (e.g. bytes, flags, rates).

Challenge E: Redundant Classification Head in MiniTransformerAutoencoder

• **Issue Description**: Model parameters were unnecessarily large, inflating the weights dictionary and communication bandwidth during federated relays.

• **Root Cause**: The model definition retained a classifier head (`self.classifier = nn.Linear(latent_dim, 2)`) that returned classification logits, a remnant of an earlier supervised hybrid design.

• **Chronological Fixes**: Commented out `self.classifier` in `__init__`, updated the forward pass to return only the `reconstructed` tensor, and cleaned up all train/eval loops to remove tuple extraction (e.g., changing `reconstructed, _ = model(X)` to `reconstructed = model(X)`).

• **Succeeded/Failed Rationale**: Streamlining the architecture succeeded because it reduced the model's footprint by removing unused linear weights, making the federated weight packet lightweight.

Challenge F: High False Alarm Rate from Strict Statistical Thresholding

• **Issue Description**: The model flagged normal traffic peaks as anomalies, resulting in a high false-positive rate and lower overall precision on the benign class.

• **Root Cause**: The dynamic threshold was computed using a strict statistical boundary of `mean + 3 * std`. While standard, network traffic has high natural variance and spikes, which easily cross a 3-standard-deviation limit.

• **Chronological Fixes**: Relaxed the dynamic boundary to `mean + 4 * std` standard deviations.

• **Succeeded/Failed Rationale**: Succeeded because the 4-standard-deviation boundary allows more headroom for normal packet spikes while remaining sensitive enough to capture massive DDoS or scan anomalies.

12. Troubleshooting Process

A rigorous debugging protocol was established to identify flaws:

1. **Syntax Validation**: Validated script modifications using python's `py_compile` module to verify syntax and imports.

2. **Feature Verification**: Printed the boolean intersection of `self.target in self.features` and the shape of `X` and `y` to ensure zero label leakage.

3. **Support Counts Inspection**: Monitored target distributions in splits to catch sequential slicing imbalances.

4. **Weight Dictionary Inspection**: Logged the keys of the weights dictionaries at the Gateway to verify aggregation inputs.

13. Tweaks and Optimizations

To adapt the simulation for constrained environments, we introduced three optimizations:

• **Weight Quantization (float16)**: Edge nodes convert model parameter weights to float16 before serialization. This reduces model parameter payload size by approximately 50% during transmission, which is then aggregated by converting back to float32.

• **Epoch Increase**: Increased training epochs from 3 to 5 to allow the local autoencoder to converge on normal traffic patterns.

• **Outlier Filtering**: Standardized outlier filtering at Edges using IQR limits to prevent training noise.

14. Final Solution

The final architecture is fully integrated in **fl_based_iot.py** and its Jupyter notebook counterpart **FL_based_IOT.ipynb**. The execution flows cleanly: data preprocessing isolates the target, stratified splits are distributed, Edges train on benign traffic, and weights are quantized to float16 and aggregated. The final model calculates a relaxed threshold and evaluates testing splits on accuracy and support-balanced reports. SHAP KernelExplainer maps anomaly contributions, saving them to a bar plot.

15. Testing and Validation

We validated the implementation using two tests:

1. **Static Verification**: Verified code compilation using `py_compile` (exit code 0).

2. **Dynamic Verification**: Ran the simulation to verify class support counts at the Gateway, Fog, and Cloud layers. All tiers confirmed balanced normal and attack samples (~8.4% attacks, ~91.6% normal).

16. Results

The classification reports demonstrate high detection metrics. Precision and recall scores for the anomaly class are non-zero and stable. In contrast to earlier iterations where accuracy was an illusion, the model successfully flags attack categories (DDoS, Ransomware, etc.) as anomalies based on reconstruction errors. This output is a textbook demonstration of the Receiver Operating Characteristic (ROC) trade-off in anomaly detection:

• **Impact of the Extra Epochs**: The local sensor training epochs were increased to 5, dropping their training losses down to 0.6425 and 0.6563. This extra optimization allowed the underlying Transformer layers to map the complex temporal dynamics of the normal IoT packets much more tightly. Because the model understands 'normal' better, the baseline reconstruction errors became more consistent.

• **Impact of Moving to 4-Sigma Threshold**: By pushing the boundary line further out to 4 standard deviations, establishing a threshold line of 0.8580, the system cleared out a massive block of false positives (dropping them to 1,504 vs. the 3,817 produced under 3-sigma). However, because network attacks can sometimes closely mimic normal traffic patterns, dropping the threshold window meant that 784 of the trickier attack packets did not create a massive enough error spike to cross the 0.8580 threshold line, exhibiting a classic sensitivity-specificity ROC trade-off.

Testing Tier	Accuracy	Anomaly Precision	Anomaly Recall	Anomaly F1-Score
Gateway Level (B.1)	84.27%	22.0%	35.0%	27.0%
Fog Level (B.2)	84.55%	23.0%	35.0%	28.0%
Cloud Level (B.3)	84.22%	23.0%	36.0%	28.0%

Table 1: Global Model Quantitative Evaluation Metrics

The relatively modest anomaly precision (22–23%) reflects the conservative reconstruction-error threshold adopted for unsupervised anomaly detection. The system prioritizes detecting previously unseen attacks (maximizing recall and security coverage) over minimizing false positives, resulting in the expected precision–recall trade-off common in anomaly detection baselines.

Testing Tier	Test Split	Normal Support	Anomaly/Attack Support	Balanced Support?
Gateway Level	B.1	13,281	1,223	Yes
Fog Level	B.2	13,280	1,223	Yes
Cloud Level	B.3	13,281	1,222	Yes

Table 2: Network Layer Anomaly Support and Split Balance

Additionally, the confusion matrices across all evaluation tiers show consistent true-positive and false-positive profiles, indicating stable boundary thresholds across the federated layers:

• **Gateway Level Confusion Matrix**: TN = 11,794, FP = 1,487, FN = 794, TP = 429

• **Fog Level Confusion Matrix**: TN = 11,834, FP = 1,446, FN = 794, TP = 429

• **Cloud Level Confusion Matrix**: TN = 11,777, FP = 1,504, FN = 784, TP = 438

Figure 2: SHAP Feature Importance Plot (Saved to SHAP_Feature_Importance.png)

17. Lessons Learned

• **Label Leakage Hazards**: Categorical fields like 'Attack_categories' contain ground truth. Machine learning pipelines must be checked to ensure no target information leaks into input feature columns.

• **The Danger of Sequential Slicing**: Shuffling and stratifying datasets is critical. Sequential splits on ordered CSVs produce skewed test subsets that invalidate evaluation metrics.

• **Asynchronous State Bugs in FL**: Gateways and intermediate aggregators must buffer weight packets in arrays rather than simple variables to avoid overwriting updates in multi-client systems.

18. Future Improvements

• **Multi-Round FL Loops**: Transition the single-round aggregation into an iterative training loop over R rounds to evaluate model convergence.

• **Secure Aggregation (SecAgg)**: Research cryptographic multi-party compute or homomorphic encryption schemas to protect parameter privacy during gateway-to-cloud transfers.

• **Adaptive Anomaly Thresholds**: Implement moving-window or node-specific dynamic threshold adjustments that adapt to drifting network flow states automatically.

• **Deployment on Edge Hardware**: Port the lightweight PyTorch model and preprocessors to real-world edge hardware (e.g. Raspberry Pi or NVIDIA Jetson) to measure inference latency and power usage.

• **Real-Time Streaming Pipelines**: Integrate real-time packet stream ingestion using message brokers (such as MQTT or Apache Kafka) for live production inference.

• **Centralized Baseline Comparison**: Run a comprehensive benchmark comparing our hierarchical federated model against a centralized learning model to verify performance tradeoffs.

• **Differential Privacy (DP)**: Inject Laplacian or Gaussian noise into local weights before upload to protect against model inversion attacks.

• **Pruning and Quantization**: Explore dynamic model pruning to further reduce communications overhead in bandwidth-constrained channels.

19. Conclusion

We successfully designed, built, and optimized a privacy-preserving hierarchical Federated Learning intrusion detection framework. By addressing critical data leakages, class support imbalances, and weight overwrite bugs, the final system represents a robust solution for IoT security. The combination of unsupervised autoencoders, quantized weight uploads (float16), relaxed anomaly thresholds, and SHAP-based feature explanations provides a practical foundation for secure, bandwidth-efficient intrusion detection.

Key Takeaways:

• Designed and built a complete hierarchical Federated Learning simulation from scratch (Edge to Gateway to Fog to Proxy to Cloud).

• Improved model reliability by eliminating target leakage and fixing distributed weight aggregation bugs.

• Demonstrated communication optimization using float16 parameter quantization during edge weight uploads.

• Integrated SHAP-based explainability for transparent reconstruction anomaly analysis.

• Evaluated the global system on the WUSTL-HDRL 2024 dataset using balanced, stratified testing splits.

20. References

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